

INBOUNDGOD project handoff

Generated 2026-10-02 from the repository `INBOUNDGOD/INBOUNDGOD`, branch `claude/keen-newton-hcb1bu`. This PDF is for a local model or a person taking over. The same content, plus every photo, demo and build file, is in the accompanying zip (`inboundgod-handoff.zip`). The zip is what a local Claude Code session should open; this PDF is the readable version.

How to continue with a local model

1. Unzip `inboundgod-handoff.zip` somewhere on your PC (or clone the repo).
2. Start Claude Code (or any coding assistant) in that folder. `CLAUDE.md` tells it to read `HANDOFF.md` first and to keep it updated.
3. If the assistant can't read files, paste the sections of this PDF you need.

Live links that still work from anywhere: watch face demo [Ultimatrix v2](#), crush site demo [Be My Crush?](#), pull request [INBOUNDGOD/INBOUNDGOD#1](#).

1. HANDOFF.md (project state)

HANDOFF

The single source of truth for picking up this project mid-stream. Any Claude session (any model) that opens this repo should read this file first, then keep it current. See `CLAUDE.md` for the update rules.

Last updated: 2026-10-02

1. Active project: Ben 10 watch face

Goal: a personal Ben 10 / Omnitrix-themed watch face for the user's Samsung Galaxy Watch.

Status (2026-10-02, later): WFS build complete on the user's PC (background, seconds bar, overlay, time, date; Always-On static + green time + minute marker). **Install to the watch: attempted, result unknown.** Run on Device didn't connect at first; the user was walked through Developer options, Wireless debugging, adb pair/connect and WFS's own "Add device" dialog (IP 192.168.0.20, pairing port 34365 at the time). The user then moved on to other projects without saying whether it worked. Also still open: the user said their WFS build "looks nothing like the v2 preview" but never sent the comparison screenshots.

Earlier status: user approved **Ultimatrix v2** and the **build kit is made** (2026-10-02): `watch-faces/ultimatrix/` (layers, fonts, previews, SVG sources and `WFS-BUILD.md`). Waiting on the user to build it in Watch Face Studio, install it on the watch and report back.

Build kit: `watch-faces/ultimatrix/`

- `WFS-BUILD.md`: the step-by-step Watch Face Studio guide (layer order, positions, fonts, colours, Always-On set, install, phone battery setup, checklist).
- `layers/*.png`: full-canvas 450 × 450 images, all placed at X 0, Y 0. `01_background`, `03_bezel_overlay`, `aod_01_static`, `aod_02_seconds_tick`. (`05_phone_icon_label` was deleted when the battery was dropped.)
- **Seconds-ring technique:** a WFS circular **progress bar** (r 202, thickness 30, #62E83C, value = seconds, range 0–60, start angle −3°) sits between the background and `03_bezel_overlay.png`. The overlay is the bezel with see-through slots, so the green bar only shows inside slots. The −3° start makes the bar stop between slots.
- Time, date and battery are WFS text, clock and complication components (not images), in Oxanium (`fonts/`, static Bold and SemiBold cut from Google's variable font with fonttools; OFL licence).
- In `aod_01_static` the hourglass lines are cut out behind the time box (x 92–358, y 166–262), because WFS text can't have the black halo the demo used.
- `render-layers.js` regenerates all images (Playwright, run from the folder).
- **Phone battery: dropped by the user.** For the record: WFS faces can't read the phone battery directly; it would need a complication slot and a phone-battery provider app.
- **Always-On seconds:** included as a seconds-hand layer for testing. Google and Samsung sources say Always-On only redraws once a minute, so it will probably freeze or be rejected by WFS. The user has been told.

Ultimatrix v2 (current design)

Live preview: <https://claude.ai/artifact/ToMBV5b9tmr8zoFQ3WkC7q> Source: [watch-faces/demos/ultimatrix-v2.html](#)

Wrist up (active): - Thick black bezel with 60 seconds segments (lit one per second), inner green rim, dark green radial background, hourglass outline. - Date (FRI · OCT 02) above; big glowing HH:MM (y 214, size 88). - **No readouts**. The user removed the left and right complications, then (while building in WFS, 2026-10-02) the phone battery too: "I don't want it anymore". The face is just time, date and the seconds ring. - The four diagonal corner notches from earlier demos were **removed**. They were decoration only, and the user asked to drop anything that's just style.

Always-On (user request 2026-10-02: "everything fades away, leave only the ticking and the Ben 10 theme"): - **Stays**: time in **solid green** (#3fbf28, light-green edge #b8ff9a), bright hourglass outline (#62e83c, 3.5 px, no fill), a thin dim-green rim ring (r 175, #2f8f22, 2 px) and **one bright seconds mark ticking round the edge** (#9dff6e). - **Brightness (user request 2026-10-02)**: the first Always-On version looked too dim, close to a blank black screen. It was brightened and measured at about **9.5% lit pixels**, under Samsung's guideline of about 15%. Filling the hourglass pushed it to 27%, so the hourglass stays outline-only. - **Goes**: date, phone battery, full seconds ring, bezel, glow. - **Seconds in Always-On: confirmed not possible in WFS** (user test, 2026-10-02). An Analog Clock in Always-On only moves the minute hand. The user says an older Ben 10 face of theirs had a "spinning circle" set to seconds; that may only have animated in the WFS preview. Also tested an image rotated by [SEC]*6: it doesn't move even in WFS's Always-On preview (which steps per minute, so [SEC] stays 0). Seconds are impossible in Always-On. **Final (user-confirmed)**: the tick image is a **Radial indicator synced to Minute in Hour** (value 0–60, 360° CW), so it orbits the rim one step per minute. - WFS reported an Always-On **On Pixel Ratio of 3.1%** for the user's build. - A black halo behind the digits stops the hourglass lines crossing them. - The demo fades over 0.9 s. On the watch, the switch to Always-On is handled by the system.

Earlier: Ultimatrix readout options (superseded by v2)

Live preview: <https://claude.ai/artifact/7xFk2CoHNcQPC4HA4m9aST> Source: [watch-faces/demos/ultimatrix-options.html](#)

Option	Left slot	Right slot
A	Weather (temp + condition)	Heart rate (BPM)
B	Next calendar event (time + title)	Sunset time
C	Rain chance	Next alarm

Recommendation given: build both slots as Watch Face Studio **complications**, with the chosen pair as defaults. The user can then swap either slot on the watch (touch and hold the face, then Customize) without a rebuild. In this layout the time moved up (y 214, size 88) to make room for an icon, value and label stack at x 175 and x 275.

Target device

Item	Value
Model	Samsung Galaxy Watch4, SM-R870 (44 mm, Bluetooth)
User-supplied code	j 7bk (meaning unconfirmed; may be a serial/batch prefix)
Screen	450 × 450 px, round, Super AMOLED
OS	Wear OS (Samsung One UI Watch). Watch Face Format (WFF) based faces

Build and install path (decided)

- The user has **Samsung Watch Face Studio (WFS)** on their PC. They build the face there and install it with WFS's **Run on device** over Wi-Fi.
- Claude's job: produce the artwork (450 × 450 layers as PNG/SVG) and a step-by-step WFS build sheet (which layer goes where, which WFS component to use for each element, tap targets, Always-On settings).
- Claude **cannot** drive WFS directly from a cloud session. It only could if the user links their computer (computer-use) in a desktop-app session.
- An APK **cannot** be built in the cloud container: `dl.google.com` (the Android SDK and Google Maven) is blocked by the network policy. Don't retry that; use WFS.

The three demos

Live preview: <https://claude.ai/artifact/4mKzh5tksjy4cCN8tLb8Dz> Source: `watch-faces/demos/omnitrix-faces.html` (self-contained HTML/SVG, all drawn in a 450 × 450 viewBox, so geometry maps 1:1 to the watch).

#	Name	What's on it	Always-On version
1	Classic Omnitrix	Grey bezel with 4 green slots; green hourglass dial; hours in left black wedge, minutes in right; date on top green, battery and steps on bottom green; green seconds dot orbiting the dark ring	Hourglass outline plus green digits only
2	Ultimatrix	Thick black ring with 60 segments lit one per second (quarter marks brighter); big glowing HH:MM; date above; battery and steps below; faint hourglass outline behind	Outlined digits, 4 quarter marks, hourglass outline
3	Omnitrix Analog	12 glowing hour slots on the bezel (larger at 12/3/6/9); hourglass dial; day in left wedge, date window in right wedge; dark hands with green stripe; red second hand (Omnitrix time-out red)	Outlined hands, 4 quarter slots, hourglass outline

Key geometry (clock degrees, 0 = 12 o'clock, clockwise, centre 225,225): - Hourglass dial radius 160 to 165; black side wedges span 52° to 128° and 232° to 308°. - Bezel ring centred at r = 206, width about 38. - Palette: green `#62e83c`, highlight `#9dff6e`, deep green `#2f8f22`, wedge black `#090c0a`, red `#ff3b30`. - Font in demos: Oxanium (Google Fonts, OFL licence, so it's fine to bundle in WFS). - Battery (82%) and steps (6,214) in the demo are placeholders. In WFS they must be bound to the real battery and step-count data.

Constraints and notes

- **Personal use only.** Ben 10 is Cartoon Network IP. The artwork is original, Omnitrix-*inspired*, and must not be published to the Galaxy Store or Google Play. Don't use official Ben 10 logos or screenshots.
- Always-On mode should stay mostly black. Samsung recommends a low lit-pixel share in AOD, so use outlines rather than filled shapes.

Next steps

1. ~~Pick a design~~ (Ultimatrix v2). ~~Make layers and build guide~~ (done).
2. **User builds in WFS following `watch-faces/ultimatrix/WFS-BUILD.md`, installs with Run on device, and reports back** (photos help): layout, seconds ring, Always-On brightness, whether the

Always-On seconds mark moves, whether phone battery shows.

3. Fix whatever comes back, by editing `render-layers.js`, re-rendering and updating the guide.
4. Later (user request): when the Forgejo move is done, make a **PDF** covering everything from this project for the user's local Claude.

History

- An earlier attempt happened in Claude Code session `session_01DzeuacYjquSzcXWwEwjdB` ("desktop-npieb66-calm-star", 2026-09-30). It stalled while waiting for a style choice, and nothing was built there.

1b. Active project: Crush site (started 2026-10-02)

Goal: a cute website where someone asks another person to be their crush. Two sides: the **asker** makes a request and gets a link; the **crush** opens it and answers yes or no; the asker sees the answer.

Decisions (user, 2026-10-02): - **Plan: demo first, then a real site.** A clickable demo of all screens is built as a claude.ai artifact to perfect the look. Then it becomes a real website anyone can open, hosted on the user's Synology NAS or a free host. - Why not a claude.ai artifact for the live version: shared data in artifacts only works for signed-in claude.ai users given access, and a crush usually won't have that. - **Vibe: "Playful game":** cute characters, a "No" button that runs away, confetti and a happy dance on yes. - Kindness rule (Claude's design choice): after a few dodges the No button stops running so the crush can genuinely say no, and the "no" screens stay gentle on both sides.

Files: `crush-site/demo.html` (the clickable demo of all three screens). Live demo: <https://claude.ai/artifact/PqofEdiz8WRLsfNHNou8SL>

Demo contents: one page with three steps: 1. **Ask:** your name, their name, an optional note, and a messenger colour (Strawberry, Sunny, Minty, Grape) → "Make my link" → a placeholder link with a Copy button. 2. **Their page:** a blob "messenger" character (hopeful/happy/sad moods) asks "Will you be [name]'s crush?". The **No** button dodges on hover/tap up to 5 times (teasing lines, Yes grows), then stays still and reads "No, sorry" so they can really decline. Yes → confetti + dance. 3. **Your answer:** status chips (link made / opened / answer). Yes → celebration; no → a gentle, kind message; none yet → waiting. - The messenger character is **interactive** (user request): it slowly drifts toward the cursor and its eyes follow it; it can be dragged and springs back on release; a click/tap squishes it and pops a heart. The "Demo" label and the demo footnote were removed (user request); the step tabs stay so the demo can be navigated. - Fonts Grandstander + Nunito; candy palette (berry #ff3d6e, sun #ffc3f, mint #4fcfa3, ink #3b1f2b on pink dots). No capabilities; state stays in memory, nothing is sent.

Status: demo published. Next: user feedback on the demo, then the real site (choose host: NAS vs free host; storage per request with a private result link for the asker).

Note: this repo is the user's public GitHub profile repo. Don't commit real names or answers from people.

1c. Active project: RTX 3060 Ti repair (started 2026-10-02)

Goal: the user's "RTX3060TI DUAL OC 8GB 256BIT 3 DP HDMI V1" artifacts heavily and the NVIDIA driver never installs properly. They suspect VRAM chip "A1" and want the board schematic/boardview to attempt a repair.

Brand confirmed: Palit (from the user's recovered ChatGPT history). The Palit Dual uses NVIDIA reference board PG142 SKU 20 (TechPowerUp). ASUS files are irrelevant. Full notes and the MODS excerpt: `gpu-repair/notes.md`. **Diagnosis so far:** MODS shows FBIOA1 with 205,298 write errors, failing bits A032–A039 = one byte lane of the FBIOA1 chip. Physical chip for FBIOA1 not yet identified (needs boardview or a freeze-spray test).

Earlier note: the name was ambiguous. "Dual OC ... V1" matches the **Palit GeForce RTX 3060 Ti Dual OC V1** exactly, but ASUS also sells a **Dual RTX 3060 Ti OC (DUAL-RTX3060TI-O8G)**. A ChatGPT chat the user shared (<https://chatgpt.com/share/6abfcd1d-9f40-83e9-a013-eb39fd9ce8ad>) assumed ASUS. Photos of the bare PCB (user promised) settle it: the PCB code is printed on the board (ASUS: CG190P / CG190PI / CG142S; Palit/Gainward: a V-number).

Files found (2026-10-02): - elvikom.pl thread "Schemat Asus RTX3060TI-8G MINI CG190P" (free after forum registration): boardviews (*.cad) for ASUS RTX3060TI-8G MINI CG190P r1.00 and DUAL-RTX3060TI-8G-MINI-I3S CG190P_HYN r1.00. <https://www.elvikom.pl/schemat-asus-rtx3060ti-8g-mini-cg190p-t73599.html?lang=en> - badcaps.net thread 3474108 (premium download): Asus_TUF_RTX3060TI_O8G_GAMING .fz + .pdf, ASUS CG142S schematic PDF + boardview PDF, Gigabyte PG190-A02 PDF, and V397_10/20/40/50/51/60/61/70 .cad boardviews. - pkbiosfix.com thread 10444 (VIP): the same file set. - realschematic.com: paid ASUS TUF CG190PI and CG142S packs. - A YouTube video claims a free download of the TUF CG190PI schematic+boardview (<https://www.youtube.com/watch?v=58wiEcPtYJU>); not verified. - No file was downloaded into the repo (paywalled/registration; not needed until the board is identified).

How "A1" maps to a chip: NVIDIA MATS/MODS reports memory by partition (FBPAA–D) and sub-partition (0/1): A0, A1, B0, B1, C0, C1, D0, D1 = the 8 GDDR6 chips on a 256-bit card. The physical position of each is board-specific; the boardview's net names (FBPA...) give the mapping. Don't guess from another board.

Repair reality (told to the user): replacing GDDR6 needs a BGA rework station (hot air/IR + preheater), a reballing stencil, a matching replacement chip (same part number, e.g. Samsung K4Z80325BC or Hynix H56C8H24AIR), and practice. Diagnose first (MATS/MODS run, rail resistance checks) before removing anything.

Status: front and back PCB photos received (shrunk copies in `gpu-repair/photos/`, layout map and plan in `gpu-repair/notes.md`). VRAM confirmed **SK hynix H56G32CS4D-X005** (8 Gb GDDR6, 180-FBGA). User has a hot-air station. User believes FBIOA1 = chip **M1** (bottom-left, above PCIe) from MATS + online research; still to be verified by a heat/freeze test. Copper debris seen between M1 and M3 must be cleaned first. Plan: clean → verify chip → optional reflow → replace with new H56G32CS4D-X005.

2. User preferences (apply to all work here)

- Likes seeing visual demos of options before committing to one.
- Uses claude.ai cloud sessions and the Claude desktop app, and switches sessions and models. Keep this file complete enough that a new session needs nothing else.
- Repo: `INBOUNDGOD/INBOUNDGOD`. Work happens on branch `claude/keen-newton-hcb1bu`; pull requests go into `main`.
- **This is the user's GitHub profile repo** (repo name = account name, description "Config files for my GitHub profile."). A `README.md` at the repo root on the default branch is shown publicly on their GitHub profile page, so don't add or change a root `README.md` unless the user asks. Project docs go in subfolders (e.g. `watch-faces/<design>/WFS-BUILD.md`).

- Open PR: <https://github.com/INBOUNDGOD/INBOUNDGOD/pull/1> (`claude/keen-newton-hcb1bu` → `main`). Every push to the branch updates it, so don't open a second PR for this work.
 - `main` was created on 2026-10-01 as an empty "Initial commit" because the repo had no base branch and PR creation failed ("no base branch ... tried 'main'"). The work branch was joined to it with a merge (`--allow-unrelated-histories`), not a rebase, so nothing was force-pushed.
 - GitHub's **default branch** was still `claude/keen-newton-hcb1bu` (the first branch ever pushed becomes the default). The user should switch it to `main` under GitHub repo Settings, then General, then Default branch. Claude has no tool to change this setting.
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3. Moving off GitHub to Forgejo on the NAS (decided 2026-10-02)

- **Decision:** the user wants all their GitHub repos moved to their own **Forgejo** server on their Synology NAS, and GitHub dropped entirely ("Move off GitHub entirely" was chosen over mirroring).
 - **Repos to move:** `INBOUNDGOD/INBOUNDGOD` (this project; it's **public** on GitHub) and `INBOUNDGOD/claude-phone-` (private, README only).
 - **Where Forgejo details live:** the user says their Forgejo setup instructions (URL, access) are in a `CLAUDE.md` on their own machine or NAS. It's not in either GitHub repo, so cloud sessions can't see it. Ask the user or read it from a local session.
 - **How (recommended):** use Forgejo's own **New Migration** → **GitHub** import, run from the NAS. It copies full history (plus PRs, issues and releases if ticked) using a GitHub token. It needs no cloud-to-NAS access and no NAS passwords in the cloud.
 - **Consequence:** Claude Code cloud sessions (`claude.ai/code`) clone from GitHub only. Once GitHub is gone, continue this project with Claude Code running locally (PC or NAS) against the Forgejo remote. `HANDOFF.md` and `CLAUDE.md` travel with the repo, so the handoff system keeps working.
 - **Status: on hold (user, 2026-10-02).** Keep working on GitHub for now. When the move is done, make a **PDF** with all the info and instructions from this project, so the user can hand it to Claude Code running locally.
 - Steps already given to the user. Not yet done. Recommended merging PR #1 into `main` first so `main` holds all the work before migrating. Recommended archiving the GitHub repos (reversible) and deleting them only after the Forgejo copies are checked.
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4. Environment notes (cloud container)

- The container is temporary. Anything not committed **and pushed** is lost.
 - Blocked: `dl.google.com` (no Android SDK, no Google Maven).
 - Available: Java, Gradle, Node, Playwright with Chromium (useful for rendering SVG to PNG and for screenshotting demos).
 - Published pages (Artifacts) survive the container and can be re-read by URL from any later session.
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5. Change log

Newest first. One line per meaningful change: date, what changed, where.

- **2026-10-02:** GPU: gave MODS download steps (levirepair image) and multimeter test points; made annotated front/back photos in `gpu-repair/photos/annotated-*.jpg`. Waiting on readings and the MODS run.
- **2026-10-02:** GPU: user has preheater, flux, kapton, no stencil. Asked about a BIOS/software bypass of the bad VRAM; told no (fuse-level, not BIOS); only a memory underclock is worth a free try. Notes updated.
- **2026-10-02:** GPU: more photos (back of PCB, chip macro). Part number confirmed H56G32CS4D-X005. Debris between M1 and M3 noted. Repair plan written in `gpu-repair/notes.md`.
- **2026-10-02:** GPU: user sent bare-PCB photos and says they ran MODS themselves; they believe FBIOA1 is chip M1. Saved shrunk photos and the chip layout map in `gpu-repair/`.
- **2026-10-02:** GPU: confirmed Palit (PG142 SKU 20). Saved the MODS excerpt and interpretation in `gpu-repair/notes.md`.
- **2026-10-02:** New project 1c: RTX 3060 Ti repair. Researched schematic and boardview sources; brand (ASUS vs Palit) still unconfirmed. Watch: recorded install attempt as result unknown. Session model switched to Fable 5.1 by the user; handoff continued without issue.
- **2026-10-02:** Crush demo: removed the demo label and footnote; made the character follow the cursor, draggable and pokeable.
- **2026-10-02:** Built and published the crush site demo (`crush-site/demo.html`). Waiting for user feedback.
- **2026-10-02:** New project: crush site (section 1b). The watch face is paused ("fine as it is for now"); the user still has to send screenshots of how their WFS build differs from the v2 preview.
- **2026-10-02:** The user found the WFS **Radial indicator** option. Synced to seconds it doesn't move in Always-On; synced to **Minute in Hour** it works. That's now the Always-On marker. `WFS-BUILD.md` AOD 3 updated. Next: Run on device.
- **2026-10-02:** The `[SEC]*6` rotation test also failed in Always-On. Seconds in Always-On are closed as not possible; the tick layer is to be deleted.
- **2026-10-02:** User built the Always-On layers in WFS (static image + green HH:MM, 3.1% on-pixel ratio). Confirmed that Always-On seconds don't move (only the minute hand). Updated `WFS-BUILD.md`. Next: Run on device.
- **2026-10-02:** User dropped the phone battery. Removed it from the WFS build (user deleting the SmallBox complication), from `WFS-BUILD.md`, `render-layers.js` and the previews (`05_phone_icon_label` layer deleted), and from the v2 demo. WFS progress: background, seconds bar, bezel overlay, time (Oxanium-Bold 88) and date (ICU date, default "Wed, Oct 28" format) done. Next: Always-On layers.
- **2026-10-02:** Learned from the user's WFS: fonts must be **installed in Windows** (WFS lists system fonts), and font settings are under **Properties**, not Style. Recorded the Add Component menu names in `WFS-BUILD.md`. The user is adding the Digital Clock now.
- **2026-10-02:** In WFS, the user has the background, seconds progress bar and bezel overlay in, and the seconds slots tick correctly in WFS's preview. Recorded the exact working progress-bar settings in `WFS-BUILD.md` (Range uses Start 357 and Angular distance 360; the preset icons break it). Next: time, date, phone icon, battery slot.
- **2026-10-02:** User started in WFS (screenshot: empty Circle project "ultimatrix", "No layers"). Their WFS version has no digital/analog template choice, so `WFS-BUILD.md` section 1 was corrected.

Walked them through adding the first layers with **+ Add Component**. Their WFS also has a **Mask** tool, a possible alternative to the bezel-overlay trick.

- **2026-10-02:** User approved Ultimatrix v2. Built the kit in `watch-faces/ultimatrix/` (layers, fonts, previews, SVG, `render-layers.js`, `WFS-BUILD.md`). Researched: phone battery needs a complication provider; Always-On seconds likely limited by the OS.
- **2026-10-02:** Brightened Ultimatrix v2's Always-On mode (solid green digits, brighter hourglass, rim ring); measured at 9.5% lit.
- **2026-10-02:** Ultimatrix v2 updated (same artifact URL, version 2): side readouts removed (phone battery only), decorative corner notches removed, ticking seconds mark added to Always-On.
- **2026-10-02:** Built Ultimatrix v2 (phone battery in the middle; Always-On fades to just the clock, hourglass and notches) in `watch-faces/demos/ultimatrix-v2.html` and published it.
- **2026-10-02:** Built and published the Ultimatrix readout options demo (`watch-faces/demos/ultimatrix-options.html`).
- **2026-10-02:** User picked the Ultimatrix design and asked to replace battery and steps. The Forgejo move is on hold; make a PDF handoff guide later.
- **2026-10-02:** User decided to move all repos off GitHub to Forgejo on their NAS. Added section 3 with the plan and its consequences.
- **2026-10-01:** Noted that this is the user's GitHub profile repo (a root `README.md` would appear on their public profile). PR #1 status: no CI configured, no reviews, mergeable.
- **2026-10-01:** User opened PR INBOUNDGOD/INBOUNDGOD#1 (<https://github.com/INBOUNDGOD/INBOUNDGOD/pull/1>) from the Claude Code UI.
- **2026-10-01:** Created the `main` branch (empty initial commit) so PRs can target it, and merged it into `claude/keen-newton-hcb1bu`. The PR from the work branch into `main` now shows all project files as additions.
- **2026-10-01:** Created this handoff system: `HANDOFF.md`, `CLAUDE.md` rules, and a Stop hook (`.claude/hooks/handoff-check.sh`) that blocks ending a turn when files changed but `HANDOFF.md` wasn't updated.
- **2026-10-01:** Built three demo faces (Classic Omnitrix, Ultimatrix, Omnitrix Analog) with Always-On previews. Published the artifact and saved the source to `watch-faces/demos/omnitrix-faces.html`.
- **2026-10-01:** Confirmed device (SM-R870, Galaxy Watch4 44 mm) and build path (Watch Face Studio). Found that APK builds are impossible in the cloud container.

2. RTX 3060 Ti repair notes

Palit RTX 3060 Ti Dual OC V1 repair notes

Card

- Palit GeForce RTX 3060 Ti Dual OC V1, 8 GB GDDR6, 256-bit, 3× DP + HDMI.
- Palit product code family: NE6306T019P2-190AD (Dual / Dual V1).
- TechPowerUp lists the Palit Dual as NVIDIA reference board **PG142 SKU 20**. The exact PCB code is printed on the board itself; confirm from photos.

Symptoms

- Heavy artifacting; NVIDIA driver never installs properly.

MODS result (excerpt recovered from the user's ChatGPT history; original TXT lost)

```
=== MEMORY ERRORS BY SUBPARTITION ===
SUBPART READ ERRORS WRITE ERRORS UNKNOWN ERRS
FBIOA0      0          32          0
FBIOA1      0        205298         0
FBIOB0      0          32          0
FBIOB1      0          32          0
FBIOC0      0          32          0
FBIOC1      0          32          0
FBIOD0      0          40          0
FBIOD1      0          40          0
```

- Failing bit range: A032–A039.

Reading it

- GA104 has 4 frame-buffer partitions (A–D), each 64-bit, split into two 32-bit sub-partitions (x0 = bits 0–31, x1 = bits 32–63). Each sub-partition is one GDDR6 chip. 8 chips × 32-bit = 256-bit.
- A032–A039 = bits 32–39 of partition A = the **first byte lane (DQ0–DQ7) of the FBIOA1 chip**. One bad byte lane in one chip, everything else fine.
- Cause is one of: that chip's DQ0–7 group (bad chip), the 8 solder balls under it (cracked joints), or those 8 traces/vias. A reflow sometimes fixes the solder case temporarily; a replacement chip is the proper fix.
- The 32/40 write errors on all other sub-partitions are a uniform baseline, likely a test artefact, not seven more bad chips.

PCB (from the user's bare-board photos, 2026-10-02; copies in [photos/](#))

- Silkscreen next to the GPU: **SP019-U**. A "PG" silkscreen sits beside the barcode sticker (B21OB009 2+00156). No "PG142" text seen; the exact Palit PCB code may be on the back (not photographed yet).
- GPU: NVIDIA **GA104-202-A1**, date code 2139A1 (SA1HKN.M3P).
- VRAM: 8 × SK hynix GDDR6 **H56G32CS4D-X005** (confirmed from a macro photo, 2026-10-02): 8 Gb, 1.35 V, 180-ball FBGA (0.75 mm pitch). Sold new on AliExpress/eBay/gpufix.de for a few USD each. Hynix "X005" memory on 3060 Ti cards has a reputation for failing with artifacts, which supports the VRAM diagnosis.
- Back of PCB: "SH14 94V-0 E248779", date code 2138; "MADE IN CHINA"; FCC/CE label. No visible damage behind M1. The "PG" silkscreen next to the sticker is cut off by the GPU stiffener frame; probably "PG142".
- VRM: 6 phases of ON Semi 3020 power stages along the bracket side (one phase pad looks unpopulated), uP9512 controller (U8001 area), inductors 1R0 2134/2136, 4R7 2128. Blue FP5K polymer caps.
- Power: single 8-pin at top-right. Fan header J15, 4-pin J10.

Memory chip layout (orientation: bracket/display outputs LEFT, PCIe

connector at the BOTTOM, 8-pin power top-right)

	[M2]	[M5]		<- above the GPU, toward the 8-pin
			[M6]	
(VRM)	[GPU]	[M7]		<- right column
			[M8]	
	[M1]	[M3]	[M4]	<- bottom row, right above the PCIe fingers

No chips on the bracket side of the GPU. Reference designators M1..M8 are printed on the PCB next to each chip.

- **User's belief: FBIOA1 = M1** (bottom-left chip, above the PCIe connector). Basis (user, 2026-10-02): MATS reported FBIOA1, and "research" said that is M1. Still not physically verified; a heat/freeze test while MODS runs is the cheap confirmation.
- **Debris:** macro photos show copper-coloured fibres/strands in the gap between M1 and M3, lying over the row of small capacitors between the two chips. Must be cleaned (IPA + soft brush) and inspected before any rework.
- Condition seen in photos: white thermal-pad residue around the GPU and memory; old paste on the die; a few small copper/orange specks between M1 and M3 along their edges (check and clean before anything else).

Which physical chip is FBIOA1?

- The user thinks M1 (see layout). Not yet verified. Needs the PG142/Palit boardview (net names FBPA_A_DQ32..39) or a physical test: run MODS and chill one chip at a time with freeze spray (or warm it with a heat gun on low); the chip whose cooling/heating changes the FBIOA1 error count is the one.

User's tools (2026-10-02)

- Hot-air station, preheater, flux, kapton tape. **No reballing stencil.** Workaround: buy new chips that ship pre-balled (most AliExpress/eBay listings do), then only pad cleanup (wick + flux) is needed; a GDDR6 180-ball 0.75 mm stencil + 0.3 mm balls is only needed to reuse a chip.

Software / BIOS bypass: asked 2026-10-02, answer: no

- There is no vBIOS for a GA104-202 that disables a memory partition. NVIDIA disables partitions by on-die fuses (e.g. GA104-150 in the 3060 12 GB), not by BIOS. A 3060 BIOS is for a different device ID; nvflash refuses it, and forcing it risks a brick and still wouldn't mask FBIOA1.
- Only software mitigation worth a try: underclock the memory heavily (MSI Afterburner, -500 to -1000 MHz) and retest. With 205k errors it will probably not be enough, but it's free.

Plan agreed in chat (2026-10-02)

1. Clean the board (IPA), remove the copper debris between M1 and M3.
2. Verify the chip: run MODS, warm M1 alone with hot air on low (~100 °C) or chill it with freeze spray; FBIOA1 count should change. If another chip changes it instead, that chip is A1.
3. Optional first attempt: reflow M1 (preheat board ~150 °C from below, flux, hot air ~380 °C with a nozzle, 60–90 s, cool undisturbed). Fixes cracked balls only; retest with MODS.
4. If still failing: replace M1 with a new H56G32CS4D-X005. Keep the pin-1 orientation of the removed chip. Needs stencil/preballed chip, flux, kapton to shield neighbours, preheater.

Multimeter test points (2026-10-02)

Annotated photos: [photos/annotated-01-front-test-points.jpg](#) and [photos/annotated-02-back-test-points.jpg](#). Card unplugged, black lead on a screw-hole ring, diode or ohm mode: 1. GPU core rail: the 6 black coils by the ON 3020 stages → 1–10 Ω to ground. 2. Blue FP5K polymer caps → measure across each; 0 Ω = dead. 3. Memory rail coils (L45/L46 by the 8-pin, L50 and L52 by Q14/Q15 near the fan header) → 20–300 Ω to ground; near 0 = shorted chip/cap. 4. 8-pin 12 V → > 1 kΩ to ground. 5. Aux coils (1R0 by U14, 1R0 and 4R7 near HDMI) → hundreds of Ω or more. 6. Small caps between M1 and M3 (after cleaning the debris) → same as the memory rail, none at 0 Ω, none cracked. 7. Caps on the back behind the GPU → inspect for cracks; ~1–10 Ω (core rail). 8. Area on the back behind M1 → inspect, measure caps. Optional powered check in MODS: memory rail coil pad ≈ 1.35 V. Probe-point arrows (red = + probe tip): [photos/probe-A-core-coils-caps-aux.jpg](#), [probe-B-memory-coil-8pin.jpg](#), [probe-C-memory-coils-L50-L52.jpg](#), [probe-D-M1-caps.jpg](#). The user's meter is manual-range (200 Ω / 2k / 20k): 200 Ω for the core coils and any "is it 0?" cap check; 2k for the memory rail, aux coils and across caps; 20k for the 8-pin 12 V pins. "1." or "OL" on the display = over range, go up one range.

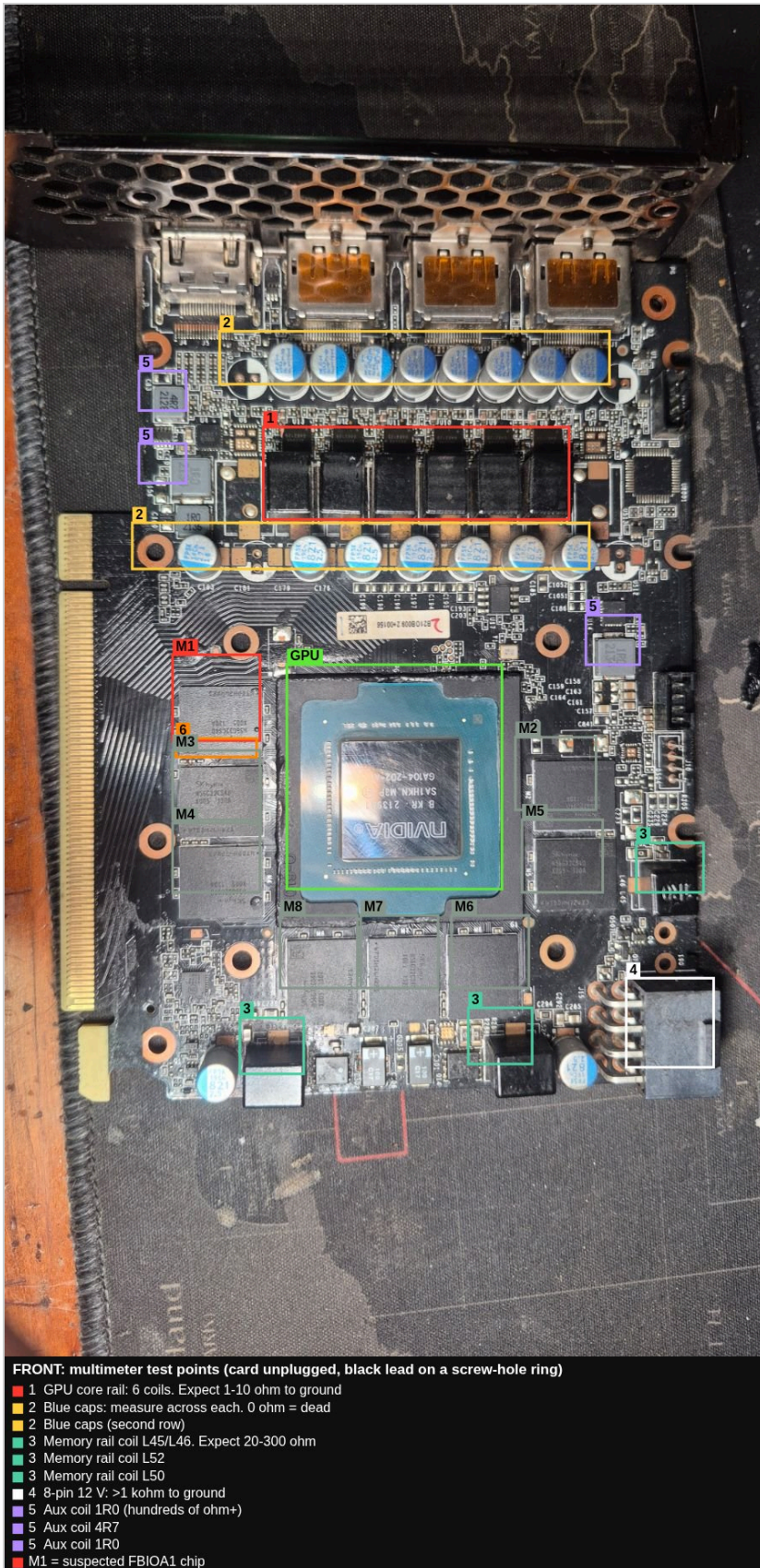
MODS/MATS source (2026-10-02)

levirepair.eu thread "Nvidia MATS & MODS USB stick (with NVMT)": 2.31 GB .7z (OneDrive + pixeldrain mirror <https://pixeldrain.com/u/ksBpKU3b>), MD5 of the image 3cc2ce1e4b9836548a8fad6479a2c4c3, needs a 32 GB stick, Rufus included. Boot → pick the 30xx entry → `mats` (card is primary) or `modsinit` then `mats -n 1` (card secondary). Use `mats -b 60 -e 70` to avoid the false 32/40 write errors on every partition (that explains the baseline in the user's report). Per-partition VRAM temperature logging: `./mods gputest.js -test 13 -run_on_error -bg_dram_temp 1000` (useful to map a warmed chip to its FBIO name).

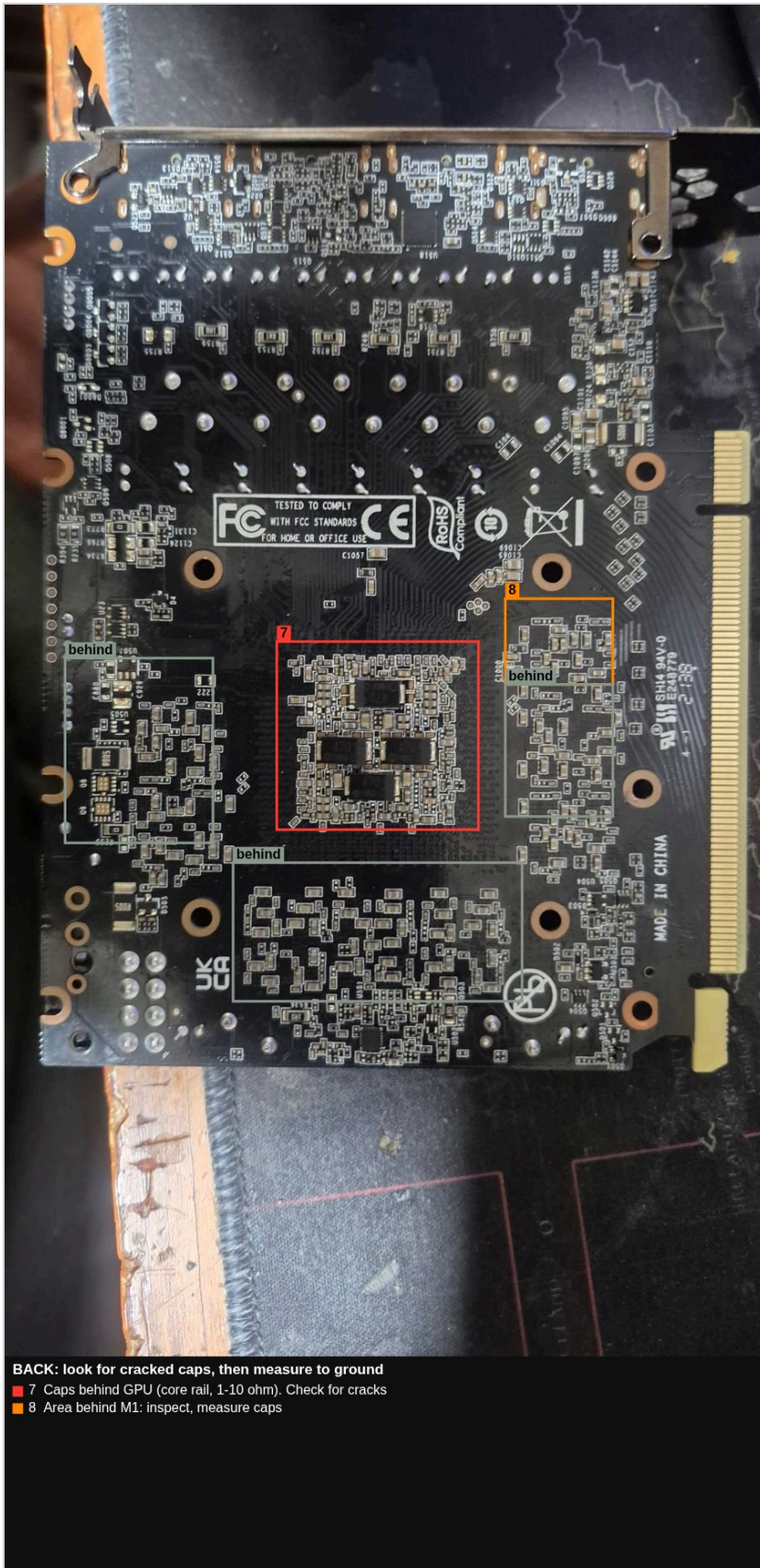
File sources checked (2026-10-02)

- badcaps.net thread 3706166 "Boardview rtx 3060ti Palit": request only, no file posted.
- badcaps.net / pkbiosfix: V397_xx.cad boardviews in the RTX 3060 packs; it is not confirmed which board "V397" is.
- Gigabyte PG142-B00 schematic exists (YouTube/Badcaps); same NVIDIA reference family, not the same board.
- Nothing downloaded yet.

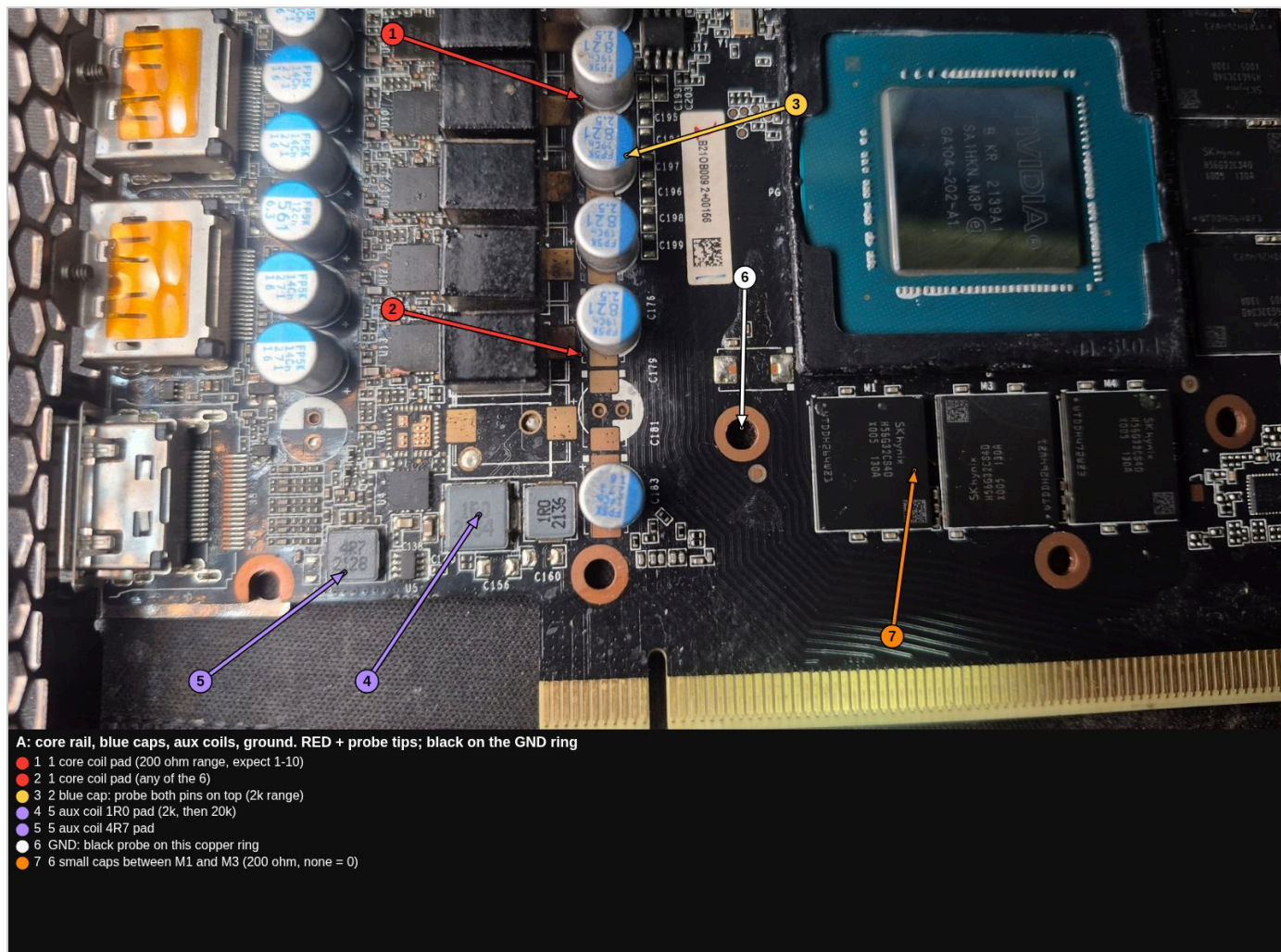
Photos



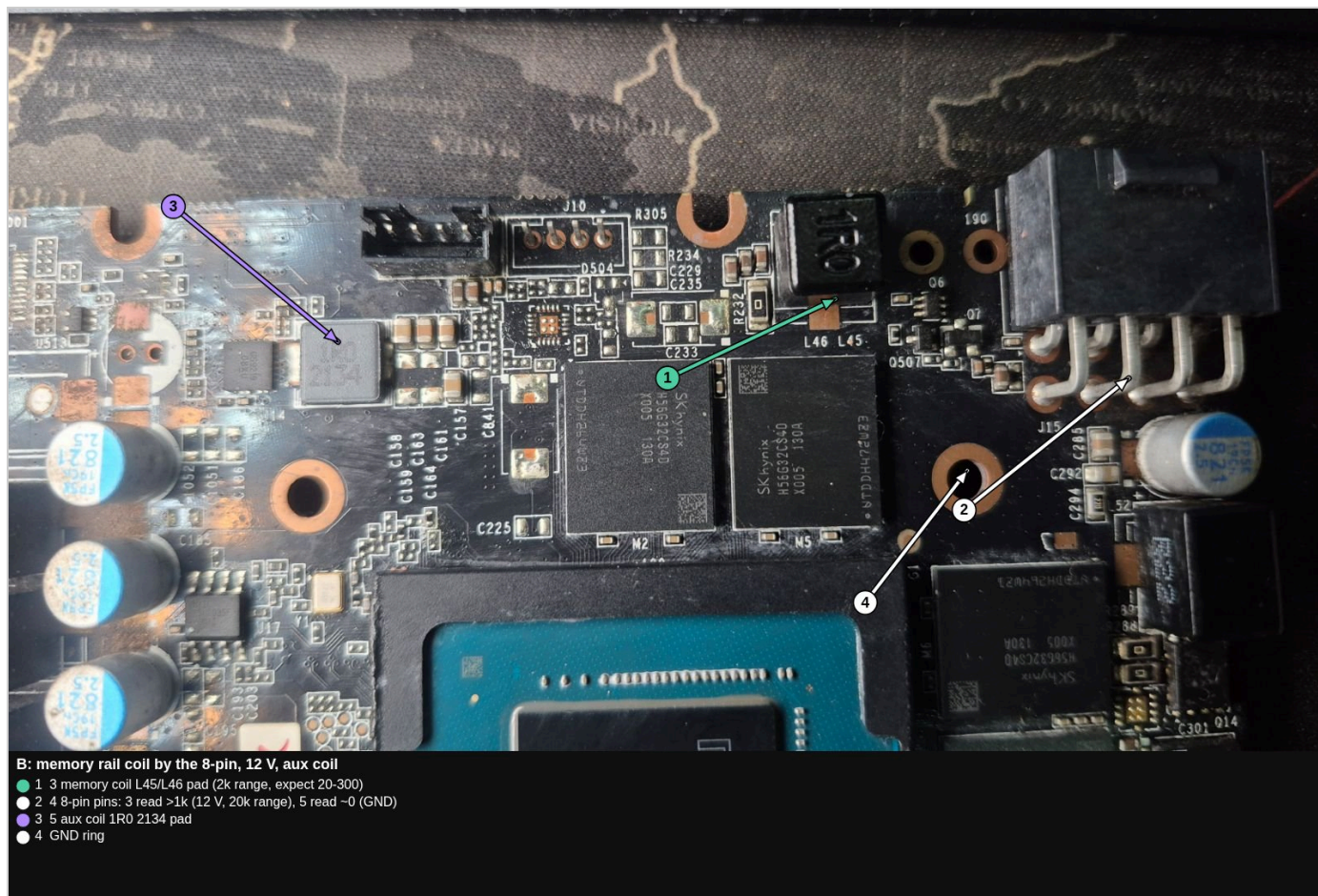
Front: test points (gpu-repair/photos/annotated-01-front-test-points.jpg)



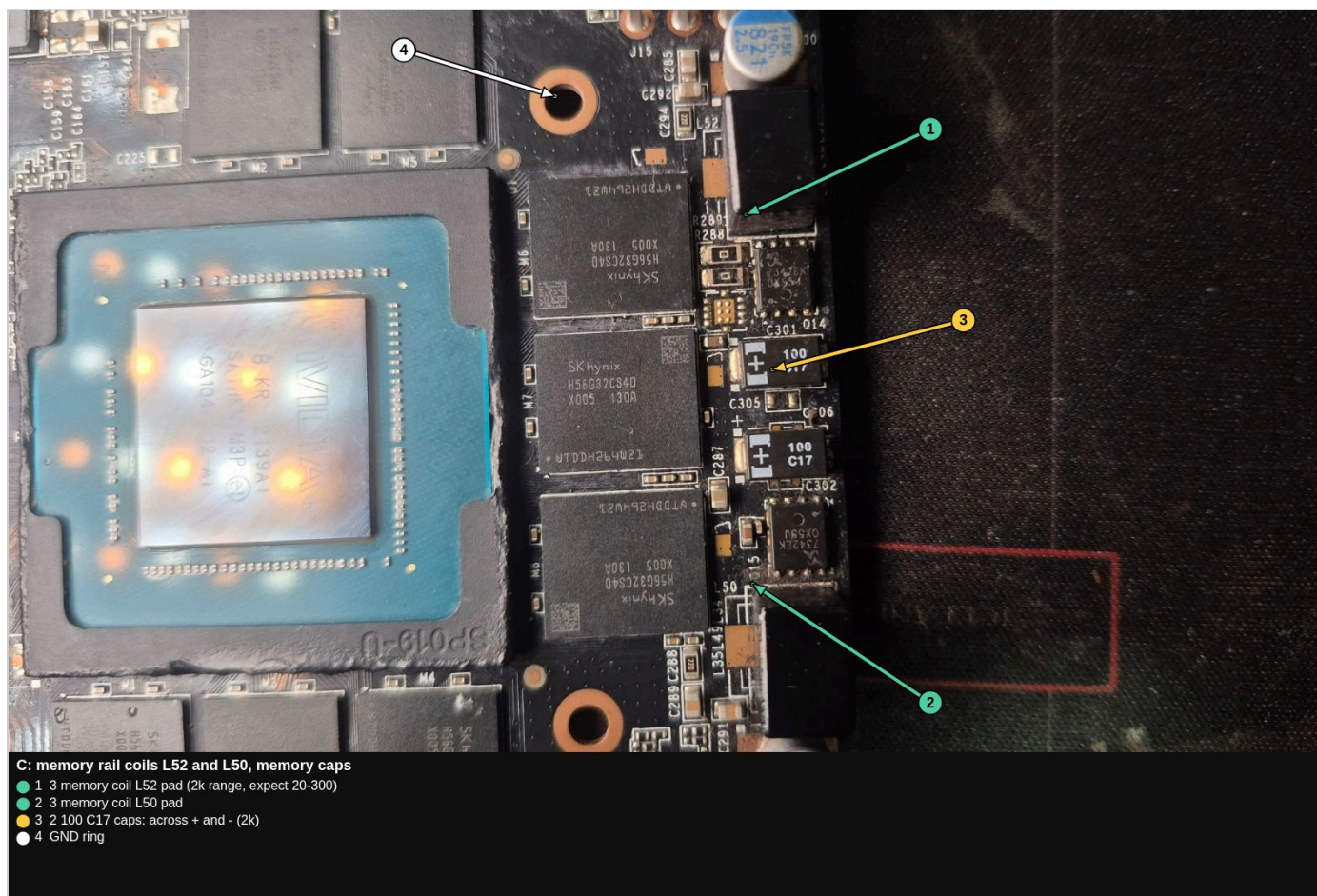
Back: test points (gpu-repair/photos/annotated-02-back-test-points.jpg)



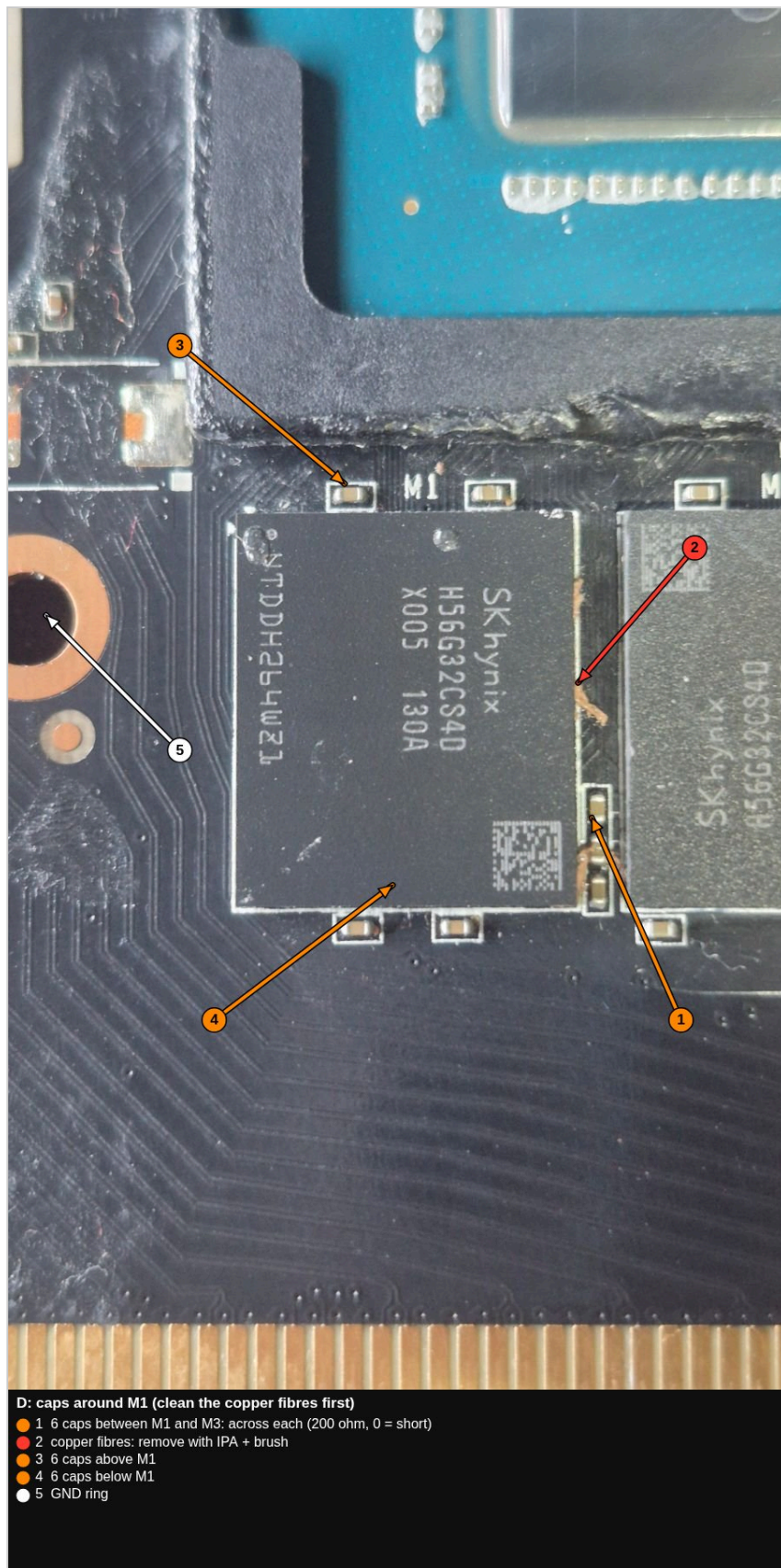
Probe A (gpu-repair/photos/probe-A-core-coils-caps-aux.jpg)



Probe B (gpu-repair/photos/probe-B-memory-coil-8pin.jpg)



Probe C (gpu-repair/photos/probe-C-memory-coils-L50-L52.jpg)



Probe D (gpu-repair/photos/probe-D-M1-caps.jpg)



M1 close-up, part number (gpu-repair/photos/17-M1-closeup-part-number-and-debris.jpg)

3. Watch face build guide (WFS-BUILD.md)

Ultimatrix watch face: Watch Face Studio build guide

Step-by-step guide for building the Ultimatrix face in Samsung **Watch Face Studio (WFS)** and installing it on a **Galaxy Watch4 (SM-R870)**.



Personal use only. Ben 10 belongs to Cartoon Network, so don't publish this face to the Galaxy Store or Google Play.

What's in this folder

File	What it is
layers/01_background.png	Everything that never moves: black face, dark bezel, unlit seconds slots, green rim, glowing centre, faint hourglass
layers/03_bezel_overlay.png	The bezel again, with see-through slots . It goes <i>above</i> the seconds bar so the green only shows inside the slots
layers/aod_01_static.png	Always-On: bright hourglass outline (with a gap behind the time) and thin green rim
layers/aod_02_seconds_tick.png	Always-On: one seconds mark pointing to 12, used as a seconds hand
fonts/Oxanium-Bold.ttf, fonts/Oxanium-SemiBold.ttf	The font (OFL licence in fonts/OFL.txt, free to use)
preview/preview_active.png, preview/preview_aod.png	What the finished face should look like. Use these to check your layout
svg/	Editable vector sources for every layer
render-layers.js	Script that regenerates layers/, svg/ and preview/

Every layer image is the full **450 × 450** size with its artwork already in the right place. Put each one at **X 0, Y 0, width 450, height 450** and it lines up on its own.

All coordinates below are on the 450 × 450 canvas, with (0, 0) at the top left and the centre at (225, 225).

1. Set up the project

1. Unzip / copy this whole `ultimatrix` folder to your PC.
2. Open Watch Face Studio → **New Project** → shape **Circle** → name it `Ultimatrix`. Current WFS versions **don't ask for digital or analog**. The project starts empty (black circle, "No layers"), and the clock is added later as a component.
3. Check the canvas is round with background **#000000**. That's the Galaxy Watch4 44 mm screen (450 × 450).
4. Everything is added with the **+ Add Component** button (top centre of the window). Each new layer appears in the **layer list on the left**, and its position, size, font and colour are set in **Properties /**

Style on the right.

5. **Install the fonts in Windows first.** WFS doesn't import fonts itself; it lists fonts installed on the PC. Right-click `fonts/Oxanium-Bold.ttf` → **Install**, same for `Oxanium-SemiBold.ttf`, then **restart WFS**. Font, size, colour and alignment are set in the **Properties** tab (text section) of a text or clock component, **not** the Style tab.
6. Menu names (current WFS): **+ Add Component** → Text, Preset image, Photo slot, Shape, Progress bar, Animation; *Time & Date*: Analog Clock, **Digital Clock ▶** (pick hour:minute), **ICU date and time** (for the date), Index, More; *Complication Slot*: Circle, Edge, Line, SmallBox, LargeBox.

Tip for exact placement: add `preview/preview_active.png` as a temporary image layer on top at about 50% opacity. Line your text up with it, then delete it before building.

2. Normal (wrist up) layers

Add these in order, bottom to top. In the WFS layer list the **top of the list is the front**, so the last one added should end up highest.

Layer 1: Background image

- **Image** component → `layers/01_background.png`
- X 0, Y 0, W 450, H 450

Layer 2: Seconds bar (lights one slot per second)

Settings confirmed working in WFS on 2026-10-02: - **Progress bar** component, Type **Circular progress bar** - Placement **X 8, Y 8**; Dimension **W 434, H 434** - Rotate **Angle 0** - Colour **#62e83c** at 100%; **Background slider 0%** (hides the grey track) - Cap style: **first (flat)** option; **Thickness 30** - Range setting: **Value [SEC]** (seconds tag), **Min 0, Max 60** - Range: **Start 357, Angular distance 360, Direction Clockwise**. Don't click the range preset icons; they turn it into a part circle and move the box. The 357 start makes the bar stop between slots, not halfway through one. - In the editor it looks like a solid green ring until Layer 3 covers it.

Layer 3: Bezel overlay

- **Image** component → `layers/03_bezel_overlay.png`
- X 0, Y 0, W 450, H 450
- Must be **above** the seconds bar in the layer list.

Layer 4: Date

- **Text** (or Date) component
- Text box: X 75, Y 130, W 300, H 26, **centred**
- Content: short weekday, then `·`, then short month, then 2-digit day. Build it with the tag picker; it should read like `FRI · OCT 02`. Use uppercase if WFS offers it; if not, `Fri · Oct 02` is fine.
- Font **Oxanium SemiBold**, size **17**, colour **#62E83C**, letter spacing about 3 (if available)

Layer 5: Time

- **Digital clock** component, format **HH:mm** (24-hour; or follow the phone's setting if you prefer)
- Text box: X 45, Y 164, W 360, H 100, **centred**; the centre of the digits should sit at about (225, 214)
- Font **Oxanium Bold**, size **88**, colour **#FFFFFF0**
- If WFS has a **glow** or **shadow** effect, add a soft green one (**#62E83C**). It's optional; the demo had a light glow.

(The phone battery slot and its icon were **removed** at the user's request on 2026-10-02. The wrist-up face is just layers 1–5.)

3. Always-On layers

Switch the editor to **Always-on** (AOD) mode with the Always-on button in the narrow icon strip next to the layer list (under the layers icon). WFS creates an automatic Always-On version; **delete everything in it** and add only these:

AOD 1: Static

- **Image** → `layers/aod_01_static.png`, X 0, Y 0, W 450, H 450

AOD 2: Time

- **Digital clock**, HH:mm, same box as Layer 5 (X 45, Y 164, W 360, H 100, centred)
- Font **Oxanium Bold**, size **88**, colour **#3FBF28** (solid green)
- No glow

AOD 3: Minute marker (orbits once an hour)

- **Image** → `layers/aod_02_seconds_tick.png`, X 0, Y 0, W 450, H 450, Inner Pivot 225, 225
- In its properties, turn on **Radial indicator** → **Apply Rotation Properties**:
- **Sync with: Minute in Hour**
- **Value: Start 0, End 60**
- **Rotate: 360°, CW**
- It steps one notch per minute around the rim, so Always-On never looks frozen.
- Why not seconds: Always-On only redraws once a minute (WFS's AOD preview runs at "1M/sec"). The user tested an Analog Clock seconds hand, a `[SEC]*6` rotation and a Radial indicator synced to seconds; none move in Always-On. Synced to **minutes**, it works (user-confirmed 2026-10-02).

WFS's own Run panel showed an **On Pixel Ratio of 3.1%** for AOD 1 + AOD 2.

Brightness check: this Always-On design lights about **9.5%** of the screen. Samsung's guideline is about 15%. If WFS warns about the Always-On pixel ratio, it should still be within the limit.

4. Install on the watch (Run on device)

1. On the watch: **Settings** → **About watch** → **Software** → **tap Software version 5 times** to unlock Developer options.

2. **Settings** → **Developer options** → turn on **ADB debugging** and **Debug over Wi-Fi**. Note the IP address it shows.
 3. Put the watch and PC on the **same Wi-Fi**.
 4. In WFS click **Run on device**, connect to the watch (enter its IP if asked) and accept the prompt on the watch.
 5. The face installs. On the watch, touch and hold the current face, swipe to **Ultimatrix** and select it.
-

5. Check it

- [] The seconds slots light up one by one and reset at the top of the minute
- [] Only the slots glow; no green shows on the solid parts of the bezel
- [] Date and time line up with `preview/preview_active.png`
- [] Lower your wrist: Always-On shows only the hourglass, rim, time (and the seconds mark, if it's allowed)
- [] Always-On is readable but not too bright

Tell Claude what you see (photos help), especially: - whether the Always-On seconds mark moves or freezes.

Regenerating the images

Needs Node.js and Playwright with Chromium:

```
cd watch-faces/ultimatrix
NODE_PATH=$(npm root -g) node render-layers.js
```

Colours, sizes and positions are at the top of `render-layers.js`.



Watch face, wrist up (watch-faces/ultimatrix/preview/preview_active.png)



Watch face, Always-On (watch-faces/ultimatrix/preview/preview_aod.png)

4. CLAUDE.md (rules for the assistant)

Instructions for Claude

Read first

Before doing anything else in this repo, read `HANDOFF.md`. It holds the current project, decisions already made, the user's preferences, environment limits and the next steps. Don't re-ask questions it already answers.

Keep HANDOFF.md current (required)

`HANDOFF.md` is how work moves between sessions and models. After every meaningful change (new files, design decisions, user answers, blockers, finished steps), update it in the same turn:

1. Edit the relevant sections (status, decisions, next steps) so they describe the project **as it is now**. Don't just append.
2. Add a dated line at the top of **Change log** saying what changed and where.
3. Update the `_Last updated:` date.
4. Commit and push, because the cloud container is temporary: `git add -A && git commit -m "..."`
`&& git push -u origin <current branch>`

A Stop hook (`.claude/hooks/handoff-check.sh`) blocks ending a turn when files changed but `HANDOFF.md` didn't. If it fires, update the file; don't work around it.

Writing rules for HANDOFF.md

- Write for a reader with zero context: full names, links, file paths, ids.
- Record **why** a decision was made, not just what.
- Never put secrets, passwords, API keys or the user's personal contact details in it.
- Keep it accurate. Delete or correct anything that's no longer true.